Driver Tuning

Pick the three driving speeds and the straight-line assist angle WITH the drivers. (issues #3 and #67, decisions 027, 031, `DriverControls.java`)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Main TeleOp RED or BLUE (gamepad 1 = driver) (TeleOp list on the Driver Station). **Needs in the config:** the 4 drive motors and the `pinpoint`

Safety: robot on blocks or on the floor with space around it, hands clear of moving parts, one person ready to press STOP. Nothing moves during INIT; everything starts after START.

Controls (gamepad 1)

Button	Does
Left stick	drive and strafe. Right stick: turn
Nothing held	NORMAL speed (70%)
HOLD a bumper	SLOW (40%), for lining up
HOLD a trigger	TURBO (100%). Bumper beats trigger
A	straight-line assist on / off. Rumble: 1 blip = ON, 2 = OFF

Course: tape a start box, a 10 ft straight run, and a 2 ft target box to park in. Each driver runs it three ways. Time it and count how many tries it takes to park inside the box.

Runs

Driver	Speed used	Assist	Time (s)	Tries to park	Felt (too fast / ok / too slow)
	SLOW	ON			
	NORMAL	ON			
	TURBO	ON			
	NORMAL	OFF			
	SLOW	ON			
	NORMAL	ON			
	TURBO	ON			
	NORMAL	OFF			

Checks

#	Do this	Expected	Actual (what happened)	If wrong
1	Drive forward with the stick a little off straight, assist ON	robot goes exactly straight		still drifts: raise `SNAP_ANGLE_DEGREES`. Can't drive a shallow diagonal: lower it
2	Strafe left with assist ON	exactly sideways, no forward creep		creeps: check `STRAFE_CORRECTION`
3	Let go of the sticks at NORMAL	robot stops, no creep		creeps: raise `DEADZONE` a little
4	Press A twice	1 blip then 2 blips, telemetry ON then OFF		

Numbers to put into the code: `opmodes/teleop/DriverControls.java`

Constant	Was	New value	Checked by
`SLOW_MODE_SCALE`	0.4		
`NORMAL_SCALE`	0.7		
`TURBO_SCALE`	1.0		
`SNAP_ANGLE_DEGREES`	15		
`ASSIST_ON_AT_START`	true		

Drivers' decision (everyone agrees): SLOW ____%, NORMAL ____%, TURBO ____%, assist starts ON / OFF. Why?

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #3 (speeds) and #67 (assist). Photos of anything surprising!